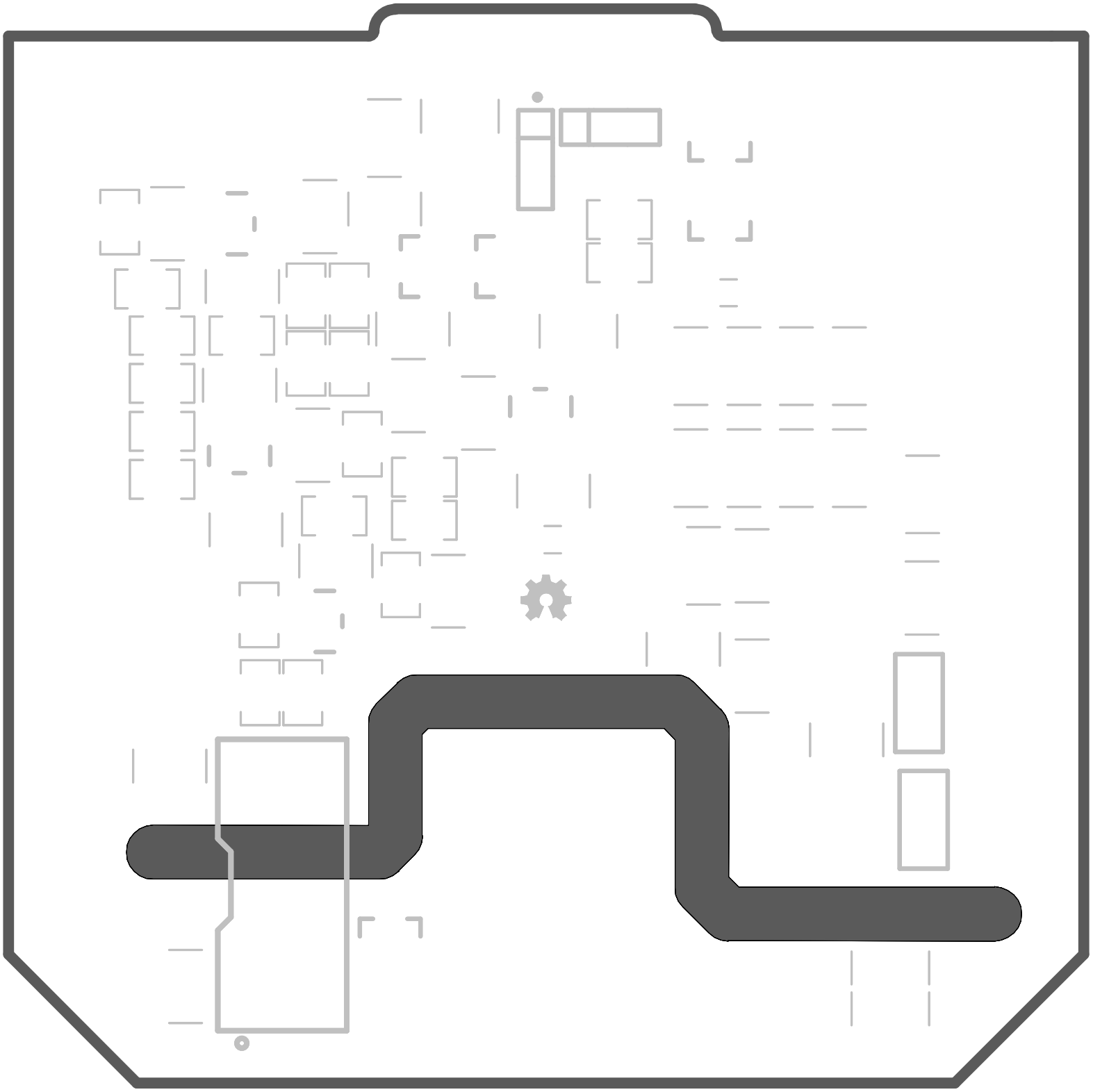
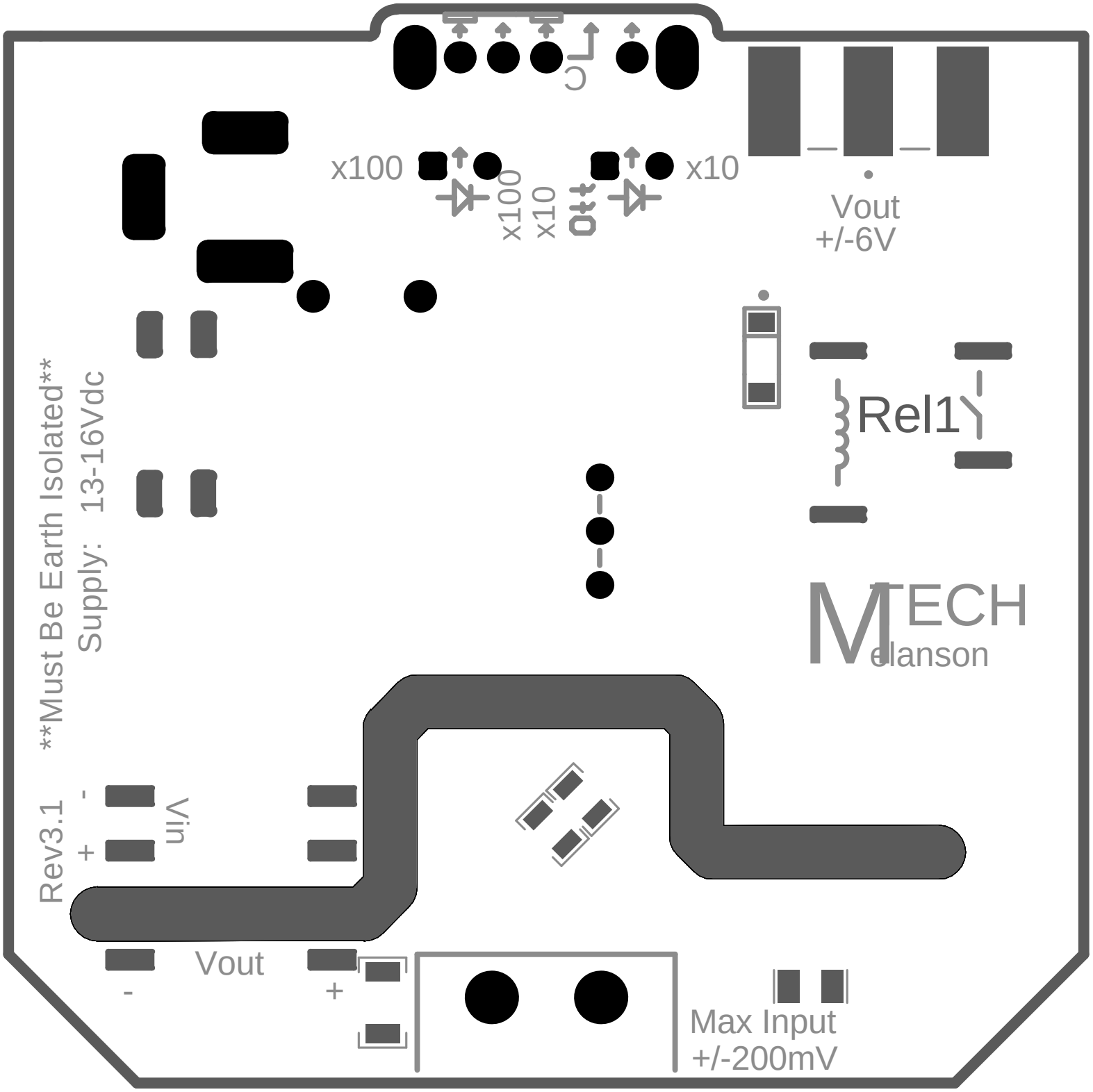
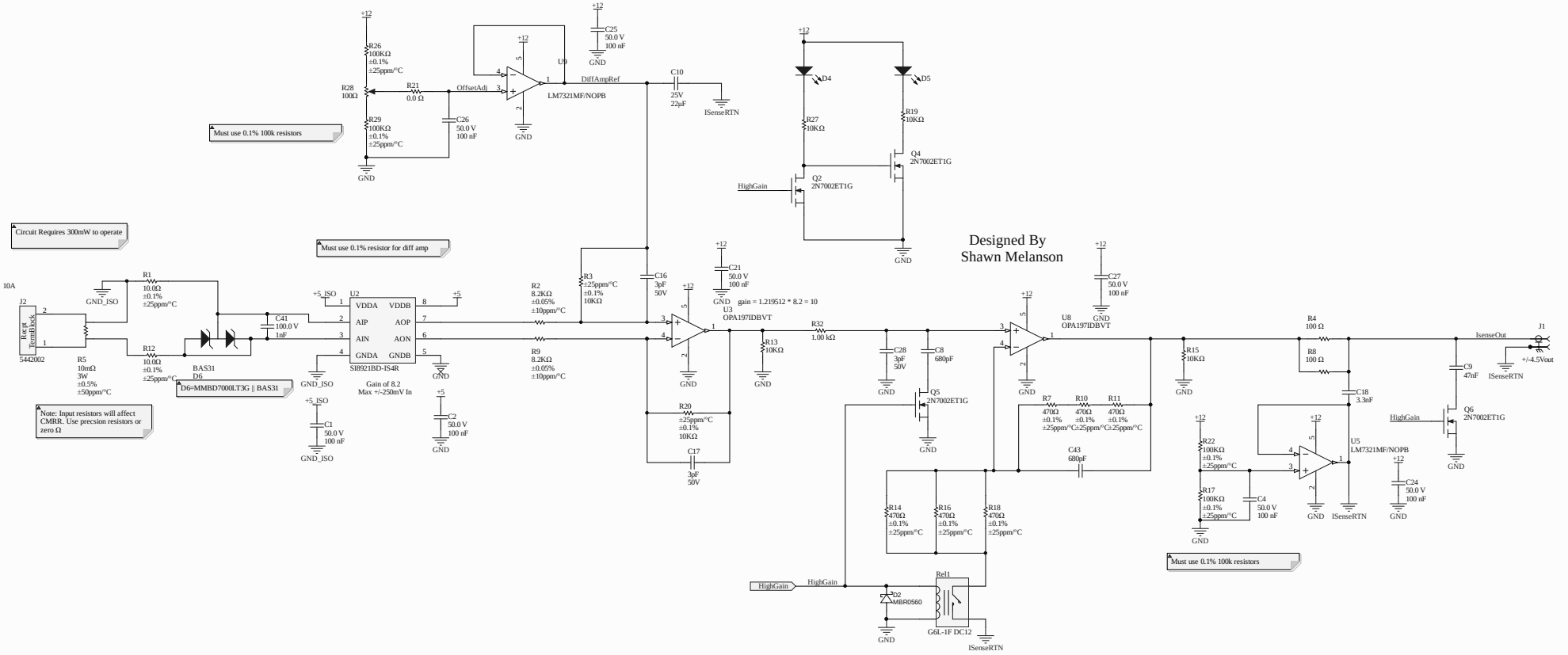


Vout
+/-6V
x10
Off
x10
x100
x100
Rail
TECH
elanson
Vin
+
-
Vout
+
-
Max Input
+/-200mV

Vout
+/-6V
x10
x100
Off
x10
x100
Rail
TECH
elanson
Vin
Vout
Max Input
+/-200mV


x100
x100
x10
x10
Vout
+/-6V
Rel1
MTECH
elanson
Must Be Earth Isolated
Supply: 13-16Vdc
Rev3.1
Vin
-
+
Vout
-
+
Max Input
+/-200mV

Isolated Shunt Amplifier

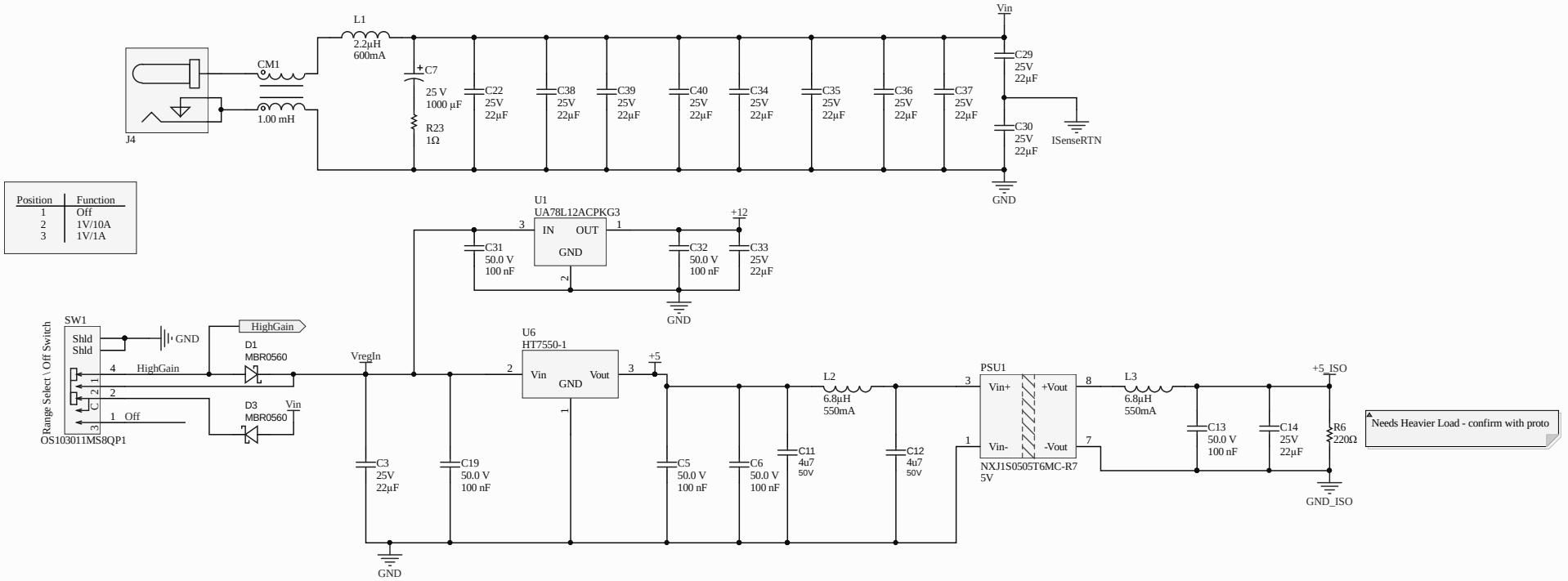


Logo1

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Title		
Size: C	Number	Revision
Date: 2026-08-20	Sheet of	
File: C:\Users\...\MelansonTech Isolated Shunt Amplifier Rev3.SchDoc	Drawn By:	

Power Supply



Designed By
Shawn Melanson

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Title		
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Date: 2026-08-20		Sheet of
File: C:\Users\..\MelansonTech PSU_IsolatedShuntAmpRev3.SchDoc		Drawn By:

Bill of Materials

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Source Data From: MelamsonTech Isolated Current Sense Rev 3.PrjPcb
Project: MelamsonTech Isolated Current Sense Rev 3.PrjPcb
Variant: None

Creation Date: 2026-08-20 8:41:32 PM
Print Date: 20-Aug-26 8:42:01 PM

Production Quantity: 1
Currency: CAD

Footprint	Comment	LibRef	Designator	JLCPCB Part #	Description	Quantity	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Order Qty 1	Supplier Subtotal 1	Manufacturer 1	Manufacturer Part Number 1
CAP, 0805, HD	CC0805KRX7R9BB104	Cap,0805,100nF,50V,5%,CC0805KRX7R9BB104	C1, C2, C4, C5, C6, C19, C21, C24, C25, C26, C27, C31, C32		CAP CER 0.1UF 50V X7R 0805	13	Digikey	311-1140-6-ND	0.10554	13	$ 1.37	Yageo Group	CC0805KRX7R9BB104
Cap,0805	CL21A226MAQNNNE	Cap,0805,22uF,25V,X5R,±20%,CL21A226MAQNNNE	C3, C14, C22, C29, C30, C33, C34, C35, C36, C37, C38, C39, C40		CAP CER 22UF 25V X5R 0805	13	Mouser	187-CL31B226KPHNNNE				Loading...	Loading...
Cap,5mm Lead Spacing,14mm HIGH,10MM Dia, Polymer - RA1	A750MV108M1EAAE014	Cap,Polar,TH,100uF,25V,Polymer,±20%,A750MV108M1EAAE014	C7		1000 µF 25 V Aluminum - Polymer Capacitors Radial, Can 14mOhm 2000 Hrs @ 105°C	1	Digi-Key	399-A750MV108M1EAAE016-ND	1.80524	1	$ 1.81	Yageo Group	A750MV108M1EAAE014
CAP, 0805, HD	Cap	CAP,0805,Generic	C8, C43			2							
CAP, 0805, HD	CL21B473KCFNNNE	Cap,0805,47nF,100V,±10%,CL21B473KCFNNNE	C9	C53134	CAP CER 0.047UF 100V X7R 0805	1	Digikey	1276-2960-1-ND	0.13886	1	$ 0.14	Samsung Electro-Mechanics	CL21B473KCFNNNE
Cap,0805	CL21A226MAYNNNE	Cap,0805,22uF,25V,X5R,±20%,CL21A226MAYNNNE	C10	C45783	CAP CER 22UF 25V X5R 0805	1	DigiKey	1276-CL21A226MAYNNNECT-ND	0.36105	1	$ 0.36	Samsung Electro-Mechanics	CL21A226MAYNNNE
Cap, 1206	CL31B475KBHNNNE	Cap,1206,4.7uF,10%,50V,CL31B475KBHNNNE	C11, C12	C26823	Cap,SMD,MLCC,4u7,50V,X7R,10%,1206	2	LCSC	C51205				Supplier disabled	Supplier disabled
Cap,0805	CC0805KRX7R9BB104	Cap,0805,100nF,50V,5%,CC0805KRX7R9BB104	C13		CAP CER 0.1UF 50V X7R 0805	1	Digikey	311-1140-6-ND	0.19441	1	$ 0.19	Yageo Group	CC0805KRX7R9BB104
CAP, 0805, HD	CC0805CRNP09BN3R0	CAP,0805,3pF,COG,NPO,CC0805CRNP09BN3R0	C16, C17, C28	C519937	CAP CER 3PF 50V C0G/NP0 0805	3	Future Electronics	9724337				Loading...	Loading...
CAP, 0805, HD	CC0805KPX7R9BB332	Cap,0805,3.3nF,50V,±10%,CC0805KPX7R9BB332	C18	C1738	CAP CER 3300PF 50V X7R 0805	1	Digikey	13-CC0805KPX7R9BB332CT-ND	0.18052	1	$ 0.18	Yageo Group	CC0805KPX7R9BB332
CAP, 0805, HD	C0805X0J221GACAUTO	Cap,0805,1nF,±5%,100Vdc,C1SF1042B8NNNC	C41	C46653	CAPACITOR, 1NF, 100V, NP0, 0805	1	Digi-Key	399-19430-1-ND	0.97205	1	$ 0.97	Yageo Group	C0805X0J221GACAUTO
CM Choke,SMD, DR331	DR331-105BE	common mode, 1mH@100kHz, 500mA, SMD,3n, DR331-105BE	CM1		CMC 1MH 500MA 2LN SMD	1							
SOD123	MBR0560	Diode,SOD123,Shottky,60V,0.5A,MBR0560	D1, D2, D3	C77338	Dio,Sig,Schtky,0.5A,60V,SOD123	3	Digi-Key	MBR0560TPMSCT-ND	0.27773	3	$ 0.83	MCC	MBR0560-TP
LED,TH,T1,3MM-Vert		LED,TH,T1,3mm,Blue,CVLB4C7	D4, D5		Blue 470nm LED Indication - Discrete 3.2V Radial	2							
SOT23-3	BAS31	Diode,Standard,Array,Series,120V,200mA,BAS31	D6	C415998	Diode Array 1 Pair Series Connection 120 V 200mA Surface Mount TO-236-3, SC-59, SOT-23-3	1	DigiKey	BAS31FSDKR-ND	0.63878	1	$ 0.64	onsemi	BAS31
CINCH_142-0701-801	142-0701-801	Conn,SMA,Female,End Launch,142-0701-801	J1		CONN SMA JACK STR 50OHM EDGE MNT	1	Digi-Key	J502-ND	8.66518	1	$ 8.67	Cinch Connectivity Solutions	142-0701-801
TERM BLOCK HDR 2POS 5.08MM,5442002	5442002	TermBlock,Header,Male,RA,TH,12A,400V,5.08mmPitch,5442002	J2		TERM BLOCK HDR 2POS 5.08MM	1	Allied	71064909		100		Phoenix Contact	5442002
Connector,Barrel,2.1mm,5.5mm,DC power Jack	PJ-002A	Connector,Barrel,ID 2.1mm,OD 5.5mm,Power Connect, PJ-002A	J4		Power Barrel Connector Jack 2.10mm ID (0.083"), 5.50mm OD (0.217") Through Hole, Right Angle	1	Digi-Key	CP-002A-ND	0.86099	1	$ 0.86	Same Sky	PJ-002A
Ind, 0805	MLZ2012M2R2HT000	Ind,0805,2.2uH,600mA,160mOhm,MLZ2012M2R2HT000	L1		FIXED IND 2.2UH 600MA 160 MOHM	1	Digi-Key	445-8658-1-ND	0.13886	1	$ 0.14	TDK	MLZ2012M2R2HT000
Ind, 0805	MLZ2012N6R8LT000	Ind,0805,6.8uH,550mA,0.25Ω,MLZ2012N6R8LT000	L2, L3		FIXED IND 6.8UH 550MA 250 MOHM	2	Digi-Key	445-6761-1-ND	0.13886	2	$ 0.28	TDK	MLZ2012N6R8LT000
Symbol,Open Source - No Text	Open Source Logo	Symbol,Open Source	Logo1		Open Source Silk Screen Logo	1							
1A SMD Module, 6 Leads,NXJ1S0505T6MC-R7	NXJ1S0505T6MC-R7	Mod,PSU,Isolated,4.5-5Vin,5Vout,1W,4.3kVrms,200mA,NXJ1S0505T6MC-R7	PSU1		Isolated Module DC DC Converter 1 Output 5V 200mA 4.5V - 5.5V Input	1	DigiKey	811-NXJ1S0505T6MC-R7CT-ND	3.7077	1	$ 3.71	Murata Power Solutions	NXJ1S0505T6MC-R7
SOT23-3	2N7002ET1G	MOSFET,N-Channel,SOT23,0.25A,60V,2N7002ET1G	Q2, Q4, Q5, Q6	C71533	Small Signal MOSFET, 60 V, 310 mA, Single, N-Channel, 3-Pin SOT-23, Pb-Free, Tape and Reel	4	Digi-Key	2N7002ET1GOSCT-ND	0.31939	4	$ 1.28	onsemi	2N7002ET1G
res, 0805, HD	RNCS0805BKE10R0	Res,0805,10.0Ω,0.1%,±25ppm/°C,0.1W,RNCS0805BKE10R0	R1, R12	C346661	RES 10 OHM 0.1% 1/10W 0805	2							
res, 0805, HD	ERA-6VRW8201V	Res,0805,8.2KΩ,±0.05%,±10ppm/°C,ERA-6VRW8201V	R2, R9	C543754	8.2 kOhms ±0.05% 0.25W, 1/4W Chip Resistor 0805 (2012 Metric) Anti-Sulfur, Automotive AEC-Q200 Thin Film	2	DigiKey	10-ERA-6VRW8201VCT-ND	0.84709	2	$ 1.69	Panasonic	ERA-6VRW8201V
res, 0805, HD	RG2012P-103-B-T5	Res,0805,10.0kΩ,0.1%,±25ppm/°C,RG2012P-103-B-T5	R3, R20	C328719	RES SMD 10K OHM 0.1% 1/8W 0805	2	Future Electronics	4000243				Loading...	Loading...
res, 0805, HD	CRCW0805100RJNEAC	Res,0805,100Ω,5%,±200ppm/°C	R4, R8	C17408	RES 100 OHM 5% 1/8W 0805	2	Digi-Key	541-100ACT-ND	0.13886	2	$ 0.28	Vishay	CRCW0805100RJNEA
Res,SMD,3637, Nonstandard,4Term	WSL3637R0100DEA	Res,3637,10mΩ,3W,±0.5%,±50ppm/°C,WSL3637R0100DEA	R5	C4188216	10 mOhms ±0.5% 3W Chip Resistor Nonstandard Anti-Sulfur, Automotive AEC-Q200, Current Sense, Moisture Resistant, Pulse Withstanding Metal Element	1							
Res, 1206	RMCF1206FT220R	Res,1206,220Ω,±1%,±100ppm/°C,0.25W,RMCF1206FT220R	R6	C4484105	220 Ohms ±1% 0.25W, 1/4W Chip Resistor 1206 (3216 Metric) Automotive AEC-Q200 Thick Film	1	Digi-Key	RMCF1206FT220RCT-ND	0.13886	1	$ 0.14	Stackpole Electronics	RMCF1206FT220R
res, 0805, HD	ERA-6AEB471V	Res,0805,470Ω,±0.1%,±25ppm/°C,ERA-6AEB471V	R7, R10, R11, R14, R16, R18	C473280	RES SMD 470 OHM 0.1% 1/8W 0805	6							
res, 0805, HD	CRCW080510K0FKTA	Res,0805,10.0kΩ,1%,100ppm,CRCW080510K0FKTA	R13, R15, R19, R27	C17414	RES SMD 10K OHM 1% 1/8W 0805	4	Digi-Key	541-3016-1-ND	0.23607	4	$ 0.94	Vishay	CRCW080510K0FKTA
res, 0805, HD	ERA-6AEB104V	Res,0805,100kΩ,±0.1%,±25ppm/°C,ERA-6AEB104V	R17, R22, R26, R29	C192074	RES SMD 100K OHM 0.1% 1/8W 0805	4	Mouser	667-ERA-6AEB104V				Loading...	Loading...
res, 0805, HD	CR0805-J/-000ELF	Res,0805,0.0Ω,150V,,CR0805-J/-000ELF	R21	C17477	RES SMD 0 OHM JUMPER 1/8W 0805	1	Digi-Key	CR0805-J/-000ELFCT-ND	0.13886	1	$ 0.14	Bourns	CR0805-J/-000ELF
res, 0805, HD	Res1	Res,0805,Generic	R23			1							
Res,TH,POT,TopAdj	CT94EW103	Res,POT,TH,10KΩ,0.5W,±10%,±100ppm/°C,CT94EW103	R28		TRIMMER 10K OHM 0.5W PC PIN TOP	1	Digi-Key	CT94EW103-ND				Loading...	Loading...
res, 0805, HD	Vishay CRCW08051K00JNEA	Res,0805,1.00MΩ,5%,200ppm,CRCW08051K00JNEA	R32	C17513	Res Thick Film 0805 1K Ohm 5% 0.125W(1/8W) ±200ppm/C Molded SMD Automotive Paper T/R	1	Digi-Key	541-1.0KADKR-ND	0.13886	1	$ 0.14	Vishay	CRCW08051K00JNEA
Relay,SMD,GS-1F 12DC	GS-1F DC12	Relay,SPST,1A,SMD,65Vdc,12Vcoil,Non Latching,GS-1F-DC12	Rly1		Telecom Relay SPST-NO (1 Form A) Surface Mount	1	Allied	70175325		1		Omron Electronic Components	GS-1F DC12
Switch,Slide,.5P3T, OS103011-Slide	OS103011MS8QP1	Switch,TH,Slide,SP3T,100MA,12V,OS103011MS8QP1	SW1		SWITCH SLIDE SP3T 100MA 12V	1	Digi-Key	CKN9543-ND	1.30533	1	$ 1.31	C&K Components	OS103011MS8QP1
SOT89-3	UA78L12ACPKG3	Reg,Linear,Pos,Fixed,12Vout,2Vin,100mA,UA78L12ACPKG3	U1	C7372179	IC REG LINEAR 12V 100MA SOT89-3	1	Digi-Key	296-50454-6-ND	0.52769	1	$ 0.53	Texas Instruments	UA78L12ACPKG3
SOIC-8 DWV	SI8921BD-IS4R	IC,CurrentSense,Isolated,8.2Gain,500Khz,SI8921BD-IS4R	U2		Isolation IC Inverters, Power Supplies, UPS 8-SOIC	1	Digikey	883-SI8921BD-IS4RCT-ND	8.7485	1	$ 8.75	Skyworks Solutions	SI8921BD-IS4R
SOT23-5	OPA197IDBVT	OPAMP, single, General Purpose,Rail-to-Rail,10MHz,SOT-23-5,OPA197IDBVT	U3, U8	C131929	General Purpose Amplifier 1 Circuit Rail-to-Rail SOT-23-5	2	LCSC	C131929				Supplier disabled	Supplier disabled
SOT23-5	LM7321MF/NOPB	OPAMP,20Mhz,MOSFET DRIVER,SOT23-5,LM7321MF	U5, U9	C128614	Single Rail-to-Rail Input/Output, ±15V, High Output Current and Unlimited Capacitive Load Operational Amplifier, 5-pin SOT-23, Pb-Free	2	Digi-Key	LM7321MF/NOPBCT-ND				Loading...	Loading...
SOT89-3	HT7550-1	Reg,Linear,Fixed,SOT-89-3,5Vin,5Vout,150mA,HT7550-1	U6	C16106	LOW DROPOUT REGULATORS(LDO) POSITIVE FIXED 55MV @ 1MA 30V 5V 150MA SOT-89(SOT-89-3) ROHS	1							
						95					$ 36.34		

Approved

Notes